

June 25, 2024

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**Re: Comments of the Vehicle Grid Integration Council to Resolution E-5326:
Approval, with modifications, of the rate design for Phase II of the Pacific
Gas and Electric Company VGI Pilots RTP rate proposed in Advice Letter
6694-E.**

Dear Sir or Madam:

Pursuant to General Order 96-B and the Comment Letter accompanying Draft Resolution E-5326 (“Draft Resolution”), the Vehicle Grid Integration Council (“VGIC”) respectfully comments on the proposed approval of the Advice Letter (“AL”) 6694-E of Pacific Gas and Electric Company (“PG&E”) submitted on September 2, 2022.

The Draft Resolution approves several rate design elements important to the launch of Phase II of PG&E’s Vehicle-Grid Integration (“VGI”) Pilots. VGIC appreciates the Commission’s continued efforts to implement Senate Bill (“SB”) 676, which directs the Commission to “establish strategies to maximize the use of feasible and cost-effective EV integration into the electric grid by January 1, 2030.”¹ As the five-year anniversary of SB 676 and the four-year anniversary of Decision (“D.”) 20-12-029 implementing SB 676 fast approach, and the 2030 deadline for maximizing VGI draws near, VGI programs remain limited in scope and scale. The Commission must accelerate support for the state’s VGI goals, particularly given rising concerns about customer energy affordability. VGIC believes the Draft Resolution is generally consistent with the 2030 goal and the needs of VGI customers and ratepayers more broadly.

VGIC generally supports the Draft Resolution and offers one recommended modification below to ensure customers have more than one choice for export compensation. VGIC does not believe our recommendation requires any delay to the planned July 11, 2024 Commission vote or the subsequent Phase II launch.

1. VGIC Recommends a Path Forward That Would Provide VGI Pilot Participants with a TOU Export Rate Option Without Compromising Pilot Goals or the Commission’s Broader RTP Initiative

¹ Public Utilities Code Section 740.16.

Resolution E-5192 directs PG&E to offer both a dynamic real-time pricing (“RTP”) option and a static time-of-use (“TOU”) export option. However, the Draft Resolution offers a different directive, stating “the proposed RTP rate can provide cost-based export compensation for the VGI Pilots customers”² and claims that PG&E did not provide sufficient evidence in the AL to authorize a TOU option at this time.

First, as a matter of process, VGIC posits that the ~20-month gap between the filing of the AL and the release of the Draft Resolution would have been enough time for PG&E to be alerted to this deficiency and subsequently address it via a Supplemental Advice Letter. VGIC is disappointed that an “absence of sufficient evidence” is cited as one of two reasons why a TOU export credit will not be authorized at this time. VGIC respectfully requests that the Energy Division staff and Commission work proactively with the investor-owned utilities in the future to address similar deficiencies, especially for suspended Advice Letters experiencing lengthy delays between submission and disposition/resolution.

The Draft Resolution offers another reason why the TOU option should not be authorized. The Draft Resolution asserts that it is reasonable for PG&E to offer only the dynamic rate option for its VGI Pilots to:

“...effectively achieve the policy goals of (1) maximizing the learnings and benefits from the ratepayer funds authorized for the VGI Pilots, and (2) ensuring that VGI Pilots customers are compensated for their exports in a manner that is aligned with the CPUC’s adopted Rate Design and Demand Flexibility Design Principles.”³

While these are worthy policy objectives that VGIC generally supports, neither D.20-12-029 nor Resolution E-5192 require customers enrolled in the VGI Pilots to be compensated for exports in a manner that is aligned with the CPUC’s Demand Flexibility Design Principles. Using the Demand Flexibility Design Principles as justification for eliminating the static TOU export rate option is inappropriate given that the Demand Flexibility Design Principles, although important, are not defined goals of the VGI Pilots in Resolution E-5192. VGIC recommends that the Commission remove this language from the Draft Resolution.

Moreover, this reference to and reliance on the Demand Flexibility Design Principles is concerning. PG&E’s VGI Pilots, which were a direct result of SB 676 and intended from inception to **maximize VGI opportunities**, seem to have been superseded by the Commission’s broader demand flexibility efforts. VGIC strongly supports aligning transportation electrification (“TE”) efforts and demand flexibility efforts and believes that demand flexibility strategies like RTP are an important element of maximizing the VGI potential. However, adopting a new framework for success and applying a new set of goals for the VGI Pilots to meet at this late stage overlooks the express purpose and importance of the VGI Pilots. The VGI Pilots were never designed to be

² Draft Resolution at 8.

³ Draft Resolution at 8.

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exclusively or even primarily an RTP experiment; they were established to spur the vehicle-to-everything (“V2X”) market and maximize VGI pursuant to SB 676.

Notwithstanding these concerns, VGIC agrees that the RTP option has the potential to benefit customers, support the grid, and help achieve the VGI Pilot goals. However, we believe there is still value in offering a TOU option, which could be attractive to certain customers. We offer additional details and a recommended path forward in the sections below.

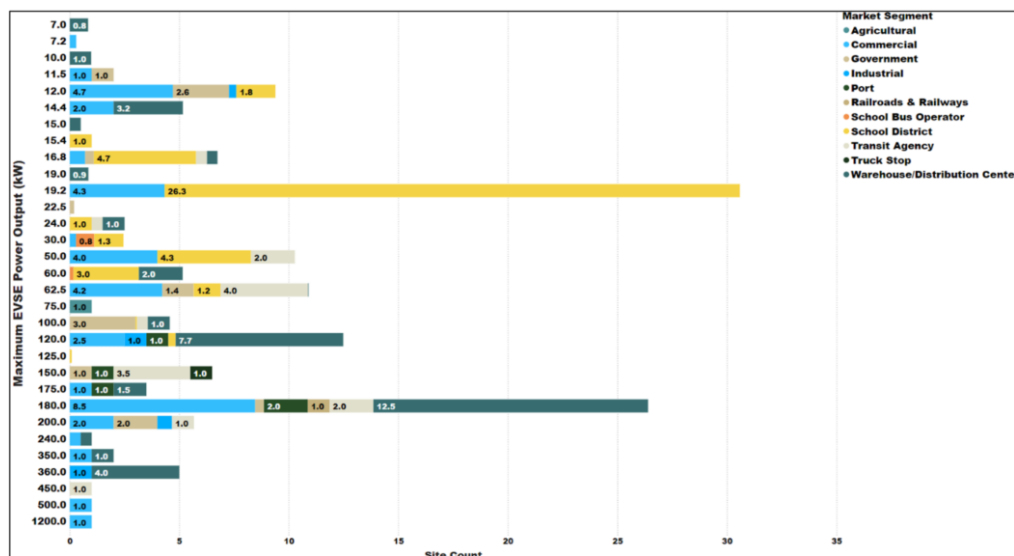
a. Customers Should be Provided a Menu of V2X Compensation Choices.

In its approval of this Draft Resolution and similar policy innovation efforts, the Commission must balance the benefits of providing clear regulatory signals to market actors and maintaining momentum on its policy initiatives with the risks of policy lock-in and path dependency toward relatively under deployed rate designs. In the case of the VGI Pilots, the Commission is “putting all its eggs in one basket” by eliminating the option of a static TOU export credit in favor of a single export compensation design.

To this end, VGIC believes that several export compensation options should be made available to customers to increase the likelihood of achieving the VGI Pilot customer enrollment goal of 1,200 customers. EV customers have a diverse set of needs and serve a variety of use cases characterized by various duty cycles, dwell times, vehicle battery sizes, charge configurations, and site load characteristics. This diversity is evident in electric vehicle (“EV”) make-ready program reporting. For example, Southern California Edison’s (“SCE”) Charge Ready Transport shows a diverse set of site host segments and charger power output:

EVSE Power Output Levels by Market Segment

D.18-05-040
D.18-01-024
A.17-01-021



Note, figures reflect active, committed sites with signed CRT Program Agreements. Partial site counts represent sites with multiple types of charging equipment.

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Figure 1. Southern California Edison, Transportation Electrification Program Advisory Council, March 8th, 2024, pg. 12 (Charge Ready Transport).⁴

This diversity was also made clear by the 2019-2020 Joint CPUC-CEC VGI Working Group, which convened over 100 representatives from utilities, state and federal agencies, ISO/RTO, automakers, charging manufacturers, charge point operators, aggregators, transit agencies, research institutions, industry associations, environmental nonprofits, consumer groups, ratepayer advocates, standards development organizations, and other key stakeholders to define, sort, assess value, and rank **2,500 potential VGI use cases**.⁵ Figure 2 below illustrates the various “dimensions” of the use case assessment:

⁴ Southern California Edison, Transportation Electrification Program Advisory Council, March 8th, 2024, pg. 12 (Charge Ready Transport).

⁵ See Final Report of the VGI Working Group. <https://gridworks.org/2020/09/final-report-of-the-california-joint-agencies-vehicle-grid-integration-vgi-working-group/>

TABLE 2. Dimensions of the Use Case Assessment Framework and Use-Case-Definition Options

SECTOR	APPLICATION	TYPE	APPROACH	RESOURCE ALIGNMENT
Residential-Single-Family Home	Customer-Bill Management	V1G	Indirect (passive)	Unified and Aligned
Residential-Single-Family Home, Rideshare	Customer-Upgrade Deferral	V2G	Direct (active)	Fragmented and Aligned
Residential-Multi-Unit Dwelling	Customer-Backup, Resiliency			Fragmented and Misaligned
Residential-Multi-Unit Dwelling Rideshare	Customer-Renewable Self-Consumption			
Commercial-Workplace	System-Grid Upgrade Deferral			
Commercial-Public, Destination	System-Backup, Resiliency			
Commercial-Public, Destination Rideshare	System-Voltage Support			
Commercial-Public, Commute	System-Day-Ahead Energy			
Commercial-Public, Commute Rideshare	System-Real-Time Energy			
Commercial-Fleet, Transit Bus	System-Renewable Integration			
Commercial-Fleet, School Bus	System-GHG Reduction			
Commercial-Fleet, Small Truck (class 3-5)	System-RA, System Capacity			
Commercial-Fleet, Large Truck (class 6-8)	System-RA, Flex Capacity			
	System-RA, Local Capacity			
	System-Frequency Regulation Up/Down			
	System-Spinning Reserve			
	System-Non-Spinning Reserve			

Given the considerable diversity of relevant use cases, it is beneficial for customers to be given the choice to participate in the option that best fits their specific needs and internal cost-benefit calculations.

In the context of the VGI Pilots Phase II, limiting available V2X compensation options solely to the RTP option, rather than offering both RTP and static TOU options – as initially directed in Resolution E-5192 – would likely unnecessarily preclude certain eligible customers from participating in the pilot. Static TOU price signals may better suit some customers' needs, goals, and technologies. Other customers may be attracted to a dynamic optimization approach that relies on RTP signals. Yet other customers may seek to not participate in either of these options and instead favor the capacity payment structure of the CEC Demand Side Grid Support: Option 3 or another non-rate program. There may not be a *one-size-fits-all* approach to unlocking V2X capability to support the grid. Just as grid planners seek fuel diversity in the bulk power generation profile, the Commission should seek V2X strategy diversification. This is especially critical when considering the relative nascency of the V2X market, where there is a need to prompt initial customer response from a wide range of new vehicle classes and use cases, customer segments, charger sizes and types, site hosts, relevant agents and actors, and price sensitivities.

By offering a portfolio of participation options, the Commission and utilities can experiment with more innovative rate designs, like the RTP approach that would be approved by the Draft Resolution, while still maintaining the key rate design principles of gradualism, simplicity, understandability, public acceptability, and feasibility of application. More widely tested export compensation mechanisms, such as static TOU export rates, can support these rate design principles. SCE appears to be developing a balanced portfolio of V2X compensation options, given its recent proposal for a static TOU export rate designed for EV customers.⁶ Notably, SCE is not proposing to exclude V2X customers from its Phase 2 Dynamic Rate Pilot (authorized in D.24-01-032), but rather offering both options and allowing customers to choose

⁶ A.24-03-019. SCE-01 at 25; SCE-04 at 121-138.

which works best for them. As detailed in Section 1.c., VGIC recommends PG&E model a TOU option after SCE's recent proposal.

b. Offering a TOU Option Would Not Interfere with the Commission's Broader Initiative to Surface Key Learnings About RTP Implementation Ahead of the CEC's 2027 Load Management Standard.

VGIC appreciates the Commission's efforts to advance RTP through the CalFUSE framework and reiterates its general support for this initiative. We believe that offering a static TOU export credit option in the VGI Pilots Phase II would not negatively impact these broader RTP efforts, as it will not meaningfully curb the lessons learned from RTP implementation. The Commission should look to the PG&E and SCE D.24-01-032 expanded dynamic rate pilots rather than the VGI Pilots Phase II for RTP implementation lessons for two reasons.

First, VGIC believes the expanded dynamic rate pilots authorized in D.24-01-032 will yield *clearer* findings about RTP implementation than the VGI Pilots because the latter are more complex. PG&E and SCE's D.24-01-032 expanded dynamic rate pilots are technology-agnostic. As a result, these pilots will see participation from customers with more established technologies than V2X charging equipment. In contrast, the VGI Pilots Phase II participants will be faced with not only a new RTP rate design but also new equipment, new upfront incentives, new mobile applications, and new relationships with vendors. These factors may make it more difficult to isolate the impact of the RTP rate design on participant behavior in the VGI Pilots compared to the D.24-01-032 pilots.

Second, the D.24-01-031 dynamic rate pilots will operate at a larger scale than the VGI Pilots. The expanded dynamic rate pilot implementation budgets exceed \$30 million and aim to achieve 50 MW of load reduction, whereas the VGI Pilots Phase II budgets are around \$10 million, pending resolution on AL 6909-E, and aim to enroll ~1,200 customers. It is reasonable to expect much clearer trends in evaluating the impact of RTP pricing on the number of customers needed to yield 50 MW of load reduction than the ~1,200 VGI Pilots Phase II customers.

For these two reasons, VGIC believes the Commission will learn more about RTP implementation from these pilots than from VGI Pilots Phase II and, as a result, establishing a TOU option should not be viewed as interfering with the Commission's broader efforts.

c. Ordering Paragraph 2 Should be Amended to Allow PG&E to Submit a Proposal to Offer a Cost-Based Static TOU Export Option to New VGI Pilot Participants by May 1, 2025.

To avoid any further delays to the launch of Phase II, VGIC recommends the Commission allow PG&E to submit an Advice Letter requesting approval for a cost-based rate design *after* the launch of Phase II. VGIC recommends PG&E develop its cost-based static TOU rate structure using relevant components from the Commission's Avoided Cost Calculator including GHG cap and trade (marginal emissions), generation energy, generation capacity, transmission, distribution,

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ancillary services procurement, and losses. This would offer a straightforward, easy-to-understand, and cost-based price signal for summer on-peak exports.

The VGI Pilots were initially expected to launch in 2022 and run for three years, and there have been several delays since the approval of Resolution E-5192. VGIC cannot state for certain whether VGI Pilots will run past 2025; however, we believe it is useful to plan for this possibility. VGIC recommends the Commission allow PG&E to propose a cost-based TOU export credit, effectively correcting the deficiencies in the AL. PG&E should aim to offer this TOU option to new VGI Pilot participants by May 1, 2025. VGIC posits that ~10 months would be a reasonable timeline for correcting the initial deficiency and subsequent CPUC disposition/resolution, given the singular focus of the effort and the foundation provided by SCE's VGRP. VGIC offers the following redline to Ordering Paragraph 2 of the Draft Resolution:

"2. PG&E is not authorized to offer a TOU option for Phase II of the PG&E Residential and Commercial Fleet Vehicle Grid Integration Pilots (VGI Pilots) at this time. **PG&E may file a Tier 2 advice letter within 90 days of the issuance of this Resolution with a cost-based static TOU export rate structure option. This advice letter shall also identify the project plan for offering the rate option to new VGI pilot participants by May 1, 2025."**

Conclusion

VGIC appreciates the opportunity to submit these comments to the Draft Resolution and looks forward to future collaboration with the Commission and other stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Interim Executive Director

Vehicle Grid Integration Council

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Service Lists: R.22-07-005 and A.20-12-011